

By definition climate data is meteorological observation data, which are more than an hour old. DMI's climate data undergoes quality control by our climatologists, through which data is checked and erroneous data is removed. In addition to the quality controlled climate data, there are several additional parameters available, which are not quality controlled, but is useful to the agricultural sector.

DMI's climate data consists of data from Denmark (DNK) and from North Atlantic Islands Faroe Islands (FRO) and Greenland (GRL). There are, however, some differences between datasets and data availability for Denmark and Greenland/Faroe Islands, respectively. Furthermore there are some key differences in how quality control is performed for Denmark and Greenland/Faroe Islands. One of the differences is that for Denmark data gaps in time and location are patched by interpolating from nearby stations, whereas for Greenland erroneous data are removed and data gaps may occur.

The Climate Data service consists of different spatial resolutions as represented in the table below.

| For climate data, there can be a delay in the updates of recent climate data over the next days

Data sets	Available for Denmark (from 2011)	Available for Greenland / Faroe Islands (from 1953)
stationValue	x	x
10kmGridValue	x	-
20kmGridValue	x	-
municipalityValue	x	-
countryValue	x	-

stationValue is quality controlled meteorological observation data for Denmark and Greenland/Faroe Islands on station level and derived values such as daily, monthly and yearly values which have been calculated for all stations. In addition to the quality controlled data, there are several additional parameters available, which are not quality controlled, but are used by the agricultural sector. For an overview of which parameters are available in the different time resolutions, please see the [parameter list](#). The stations available for the stationValue service can be seen [here](#). stationValue is available from a stations start date until the termination date of the station.

There are some key differences between stationValue for Denmark and Greenland/Faroe Islands respectively, as shown below:

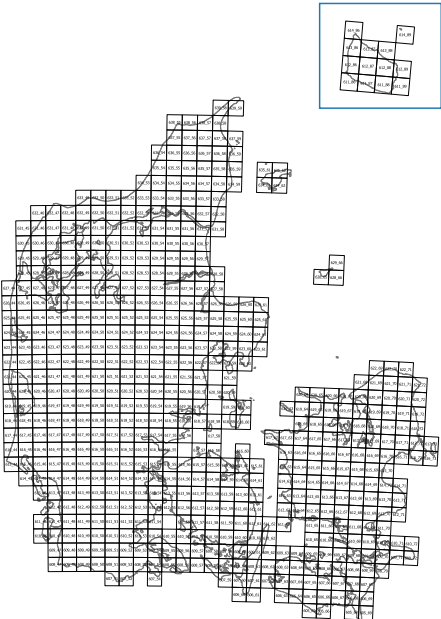
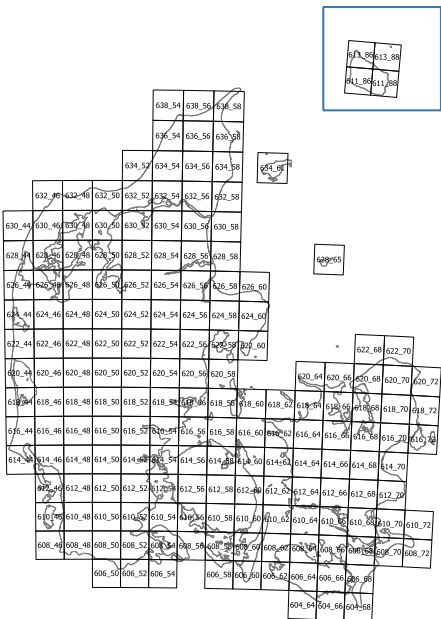
	Denmark	Greenland/Faroe Islands
Frequency of manual quality control	Daily*, monthly, yearly	Monthly
Method	Visual control by inspection of interpolated grid maps on which the station data plotted. Visual inspection of station series time plots.	At the beginning of every month, all data from the previous month is manually checked from time series plots
Complete time series	Yes Interpolation algorithm is applied to fill in data gaps and replace erroneous data	No Erronuous data is removed, an interpolation algorithm is not applied on the Greenland/Faroe Islands data, due to the complicated circumstances of the large geographical area, complex topography and low station density. In certain cases erroneous data is replaced by a manually estimated value

* The manual daily QC entails that data usually is available on the next week day after observation time.

The climateData service also includes data in different spatial resolutions for Denmark. The data is based on station data from Danish stations, which is interpolated into a 1x1km grid net covering all of Denmark's land area using a modified inverse-distance algorithm. The calculated values in each grid cell depends on the values of the nearest surrounding station found in 8 sectors. One value is calculated for each area (e.g. one municipality) from the grid cells covering the jurisdiction area combined with a weighting of how much of each grid cell is within the area. The stations are weighted in relation to the distance and climatological comparability (distance to sea) to the grid cell. Aggregated grid cells are used to compile datasets of different spatial resolution.

The different spatial resolutions are calculated/derived from stationValue data as it becomes available. The data is recalculated when the quality controlled stationValue data has been quality controlled by our climatologists. In some instances area data is based on additional parameters, which are not quality controlled. Please see the [parameter list](#) for an overview over which parameters are available and whether or not they are quality controlled by our climatologists.

Spatial resolutions are not available for Greenland, as an interpolation algorithm cannot be applied due to the complicated circumstances of the large geographical area, complex topography and low station density.

Spatial resolution	Description
10kmGridValue	Climate data in a 10 x 10km grid resolution covering Danish land areas. For your convenience DMI has created an interactive web page map to inspect the grid here (external site). You can also see, how you can import grid identification layers into QGIS. 10 km grid cells are also available as pdf here. 
20kmGridValue	Climate data in a 20 x 20km grid resolution covering Danish land areas. For your convenience DMI has created an interactive web page map to inspect the grid here (external site). You can also see, how you can import here. 
municipalityValue	climate data for the Danish municipalities

Spatial resolution	Description
countryValue	climate data for Denmark

Read more about the spatial resolutions [here](#).

A climatological standard normal is the average of a 30-year period of weather data calculated for every month of the year. A climatological standard normal contains information about the expected mean air temperature for January in Denmark and thus gives you an understanding of the weather as it “normally” is.

Climatological standard normals are widely used e.g. within climate monitoring where it serves as a reference to compare weather observations with. When climatological standard normals are compared to current weather observations they help you understand whether the mean air temperature is higher or lower than average.

Climatological standard normals are calculated on the basis of quality-controlled station data. In general, the calculations can be divided into four groups; averages, extreme values, accumulated values and event counts. For a detailed description of the climate normal calculation methods, please read [WMO Report 1203](#) (external site).

Regarding Greenland

In certain cases, the data behind the climate normal is not sufficient due to missing data in the climate normal period. In such cases the full climate normal from e.g. 1991-2020 cannot be calculated. Instead of leaving out valuable data from stations that do not have the full data coverage from 1991-2020 shorter averages are calculated:

- Period averages: Averages of climatological data computed for any period of at least ten years starting on 1 January of a year ending with the digit 1 (WMO Report 1203).
- Averages: The mean of monthly values of climatological data over any specified period of time (not necessarily starting in a year ending with the digit 1) (WMO Report 1203).

At least ten years of data has been prioritized, but in certain cases e.g. due to unique locations averages has been calculated for fewer years.

Technically, the calculation of a reference value is the same as with a climatological standard normal; however, the period it covers is shorter than the 30-year period that is characteristic for the climatological standard normal values. The reference values normally cover a period of 10 years, but the period can be both longer and shorter. A reference value thus contains an average of e.g. the mean temperature for each month during the year calculated on the basis of a number of year's data for the month in question e.g. the average temperature for January in Denmark.

The reference values are used in different contexts. One example is to illustrate how the weather in the last decade was, compared to the actual weather observations.

10-year reference values will not have the same climatological weight as the 30 years climatological standard normals, since weather related coincidences could affect the calculations.

A reference value is calculated on the basis of quality-controlled station data. In general, the calculations can be divided into calculations of the average, extreme values, accumulated values and event counts.

For a more detailed description of the calculations of climatological standard normals, see the [WMO-report 1203](#) (external site).

Meteorological observation data may contain various errors and data outages. When potential errors are identified they are investigated closely. For example by looking at raw observation data or including other data sources such as nearby stations or other parameters which can help to conclude whether the data must be excluded. Errors are typically due to malfunction of instruments caused by wear and tear or exposure to weather and on rare occasions vandalism.

Typical errors in the observations are unrealistically high or low outliers or small but consistent biases. Furthermore certain stations can be misleading for longer periods, if the station observes data deemed unrepresentative by DMI's climatologists.

stationValue for Denmark undergoes manual quality control on multiple levels:

- Spatial controls on a daily, monthly and yearly level are performed by visually inspecting interpolated grid maps on which the station data plotted
- Visual control of the station series are performed based on time plots

Erroneous or missing data values are filled in by interpolated values derived from surrounding data.

The manual daily QC entails that data usually is available on the next week day after observation time.

Climate data from DMI's open data is quality controlled daily, monthly and annually, and is usually ready in February. Unfortunately, the quality control status (qcstatus) does not indicate whether or not the data has undergone the yearly quality control. Release notes for the climate data service will be updated, when data has undergone the yearly quality control.

For waste management data packs (danish: deponipakker), DMI recommends to first download the data after it has undergone the annual quality control. Waste management data packs (danish: deponipakker) usually include wind direction, wind speed, mean temperature, precipitation, potentiel evaporation, and relative humidity.

stationValue for Greenland/Faroe Islands undergoes manual quality control to ensure the data quality:

- At the beginning of every month, all data from the previous month is manually checked from a time series plot
- Identified erroneous data will be labeled as excluded (see validity property [here](#))
- In certain cases erroneous data is replaced by a manually estimated value

Excluded values are not included in derived values. Derived values contain information on the amount of data behind the calculations.

Station values are available in the following time resolutions:

- Hour
- Day
- Month
- Year
- 10 years (also called reference values)
- 30 years (also called climate normals)

Station values are derived from meteorological observations and represent a parameters value in a specific time interval. For example, the mean temperature of the day November 3rd 2012 for a given station.

However, the exact time a certain time resolution begins and ends depends on the time zone, and how the time resolutions above are defined. Due to historical reasons the time resolutions can differ for the different parameters. Time resolutions also differ between stations values for Denmark and Greenland/Faroe Islands

The response properties **from** and **to** will tell you the exact time interval covered by the station value. If you want to query for station values in a specific time interval, you will need to know how time resolutions and types of days work.

Hourly station values are always in UTC. This avoids any confusion that might arise if summer- and winter time was used. The API supports querying time ranges with time zones, so no special knowledge is needed for querying hourly values in a specific time range.

Daily station values operate with different definitions of what a day is. The different definitions can be seen below.

Type of day	Description
Local day	from/to midnight in local time with changes to summer and winter time
Old meteorological day	a day which covers the hours from 6:00 am to 6:00 am next day in UTC
Manual precipitation day	a day which covers the hours from 12:00 pm to 12:00 pm next day in UTC. It is only used by a few stations on Greenland, where precipitation data is manually measured.
The exeption to the rule	only used by the parameter of maximum precipitation in 30 minute intervals. This is defined as the local day, but also includes the 30 minutes before midnight, so the maximum precipitation at midnight is included.

The type of day used for a station value, depends on the parameter of a station value. You can see a list of which parameters use which type of day [here](#).

Monthly station values are derived from daily station values. They begin when the first day of the month begins, and end when the last day of the month ends.

Example: The monthly station value for the parameter mean_temp in March 2018 for a Danish station would represent the time range: 1/3/2018 00:01 CET - 31/3/2018 24:00 CEST. Notice the change to summertime, which happens in Denmark on the last Sunday of March.

Even if a parameter has monthly station values but not daily values ([as per the parameter table](#)), the month still begins and ends at the time as the first and last day would have begun and ended if they existed.

Yearly station values are derived from daily station values. They begin when the first day in the year begins (January 1st), and end when the last day in the year ends (December 31st).

Example: The yearly station value for the parameter mean_temp in 2018 for a Danish station would represent the time range: 1/1/2018 00:01 CET - 31/12/2018 24:00 CET.

Even if a parameter has yearly station values but not daily values ([as per the parameter table](#)), the year still begins and ends at the time the first and last day would have begun and ended if they existed.

Municipality values are derived from daily municipality values for Climatological reference values (see below). for the period 2011-2020The specific timerange for the data collection is:

*the time range: 1/1/2011 00:01 CET - 31/12/2020 24:00 CET.*and exist for danish municipalities only.

Note, that due to the statistical nature of the data and the very long time period, the period is just described as:

2011-01-01T00:00:00+00:00 to 2020-12-31T23:59:59+00:00

Station values are derived from daily station values for Climatological normal values (see below), which at present the two 30 year periods

1961 - 19901991 - 2020and exist for stations on Greenland and the Faroe Islands only (GRL and FRO).

The specific timeranges for the data collections, in UTC, are:

01/01/1960 06:00 - 01/01/1991 06:00 for GRL and FRO

01/01/1991 06:00 - 01/01/2021 06:00 for GRL

01/01/1991 00:00 - 31/12/2020 24:00 for FRO

Note, that due to the statistical nature of the data and the very long time period, the period is just describes as:

1961-01-01T00:00:00+00:00 to 1990-12-31T23:59:59+00:001991-01-01T00:00:00+00:00 to 2020-12-31T23:59:59+00:00

Some of the parameters can assume special values which are not to be taken literally, but are codes for special cases or conditions. The codes currently in the system are listed below.

The parameters acc_precip, max_precip_past24h, acc_precip_past12h, acc_precip_past24h and max_precip_30m return the amount of precipitation within the given period in mm with the exception of the codes listed below.

Code	Value
-0.1	traces of precipitation, less than 0.1 kg/m2

The parameters mean_wind_dir & mean_wind_dir_min0 return the direction from which the wind blows in degrees, where north is given as 360.

Code	Value
0	calm

The parameter mean_cloud_cover is reported in oktas (eights) and is the fraction of the sky covered by cloud of any type or height above the ground.

Code	Value
0	0 oktas - no clouds
10	1 okta or less, but not zero
25	2 oktas
40	3 oktas

Code	Value
50	4 oktas
60	5 oktas
75	6 oktas
90	7 oktas or more, but not 8 oktas
100	8 oktas - full cloud cover with no breaks
112	Sky obscured by fog and/or other meteorological phenomena

On this page you can find a table of parameters for which climate data values are available for download through the climate data service. The calculation methods for the stationValue parameter values are described in a [separate table](#).

The available parameters, the parameter and whether our climatologists perform quality control (QC) for the respective parameters. Click [here](#) to see which resolutions data are available.

Name	Unit	Description	QC performed by DMI's climatologists
mean_temp	°C	Mean temperature	x
mean_daily_max_temp	°C	Mean of daily maximum temperature	x
max_temp_w_date	°C	Maximum temperature with associated date	x
max_temp_12h	°C	Maximum temperature within the last 12 hours	x
no_ice_days	no. of days	Number of ice days (maximum temperature < 0°C)	x
no_summer_days	no. of days	Number of summer days (maximum temperature > 25°C)	x
mean_daily_min_temp	°C	Mean of daily minimum temperature	x
min_temp	°C	Minimum temperature	x
min_temperature_12h	°C	Minimum temperature within 12 hours	x
no_cold_days	no. of days	Number of cold days (minimum temperature < -10 °C)	x
no_frost_days	no. of days	Number of days with frost (minimum temperature < 0°C)	x
no_tropical_nights	no. of days	Number of tropical nights (minimum temperature > 20°C)	x
acc_heating_degree_days_17	no. of days	Accumulated heating degree days (17°C - mean_temp) (Danish: graddage - Danmark)	x
acc_heating_degree_days_19	no. of days	Accumulated heating degree days (19°C - mean_temp) (Danish: graddage - Grønland)	x
mean_relative_hum	%	Mean relative humidity	x
max_relative_hum	%	Maximum relative humidity	x
min_relative_hum	%	Minimum relative humidity	x
mean_vapour_pressure	hPa	Mean vapour pressure	x
mean_absolute_humidity	g/m ³	Mean absolute humidity	x
mean_wind_speed	m/s	Mean wind speed	x

Name	Unit	Description	QC performed by DMI's climatologists
max_wind_speed_10min	m/s	Maximum wind speed (10 minutes average)	x
max_wind_speed_3sec	m/s	Maximum wind speed (3 seconds average)	x
no_windy_days	no. of days	Number of windy days (mean wind speed $\geq 10.8\text{m/s}$)	x
no_stormy_days	no. of days	Number of stormy days (mean wind speed $\geq 20.8\text{m/s}$)	x
no_days_w_storm	no. of days	Number of days with storm (mean wind speed $\geq 24.5\text{m/s}$)	x
no_days_w_hurricane	no. of days	Number of days with hurricane (mean wind speed $\geq 28.5\text{m/s}$)	x
mean_wind_dir_min0	degrees	Mean wind direction (10 minutes average) at minute 0	x
mean_wind_dir	degrees	Mean wind direction	x
mean_pressure	hPa	Mean pressure	x
max_pressure	hPa	Maximum pressure	x
min_pressure	hPa	Minimum pressure	x
bright_sunshine^	minutes/hours	Minutes/hours of bright sunshine. For hourly time resolution the unit is minutes, for all other time resolutions the unit is hours.	x
mean_radiation	W/m ² / MJ/m ²	Mean radiation (spectral range: 305-2800nm). For hourly time resolution the unit is W/m ² , for all other time resolutions the unit is MJ/m ² .	x
acc_precip	mm	Accumulated precipitation	x
max_precip_24h	mm	Maximum 24-hour precipitation with associated date	x
acc_precip_past12h	mm	Accumulated precipitation during the past 12 hours	x
no_days_acc_precip_01	no. of days	Number of days with accumulated precipitation $\geq 0.1\text{mm}$	x
no_days_acc_precip_1	no. of days	Number of days with accumulated precipitation $\geq 1\text{mm}$	x
no_days_acc_precip_10	no. of days	Number of days with accumulated precipitation $\geq 10\text{mm}$	x
no_days_snowfall	no. of days	Number of days with accumulated precipitation $\geq 0.1\text{mm}$	-
acc_precip_past24h	mm	Accumulated precipitation during the past 24 hours	x
max_precip_30m	mm	Maximum 30 minutes intensity in 24 hours with date	x
no_days_snow_cover	no. of days	Number of days with snow cover (> 50% covered)	x
no_days_fog	no. of days	No. of days with fog (vis. < 1 km)	-
mean_cloud_cover	%	The fraction of the sky covered by cloud of any type or height above the ground (see also Codes)	x
no_clear_days	no. of days	Number of clear days (mean cloud cover < 20%)	x

Name	Unit	Description	QC performed by DMI's climatologists
no_cloudy_days	no. of days	Number of cloudy days (mean cloud cover > 80%)	x
max_snow_depth	cm	Maximum snow depth with date and location	x
snow_depth	cm	Snow depth	x
snow_cover	%	Snow cover	x
temp_grass *	°C	Air temperature measured at grass height (5-20 cm over terrain)	-
temp_soil_10 *	°C	Temperature measured at a depth of 10 cm	-
temp_soil_30 *	°C	Temperature measured at a depth of 30 cm	-
leaf_moisture *	minutes	Leave moisture	-
vapour_pressure_deficit_mean *	kPa	Vapour Pressure Deficit Mean	-

^For bright_sunshine, the algorithm to calculate hours of sunshine has been changed since week 11 2026.

* The last 5 parameters in the table above are additional parameters, which are not quality controlled by DMI's climatologists. As a result, data can be erroneous. ** Read more about the different types of days [here](#).

*** 30 years reference values, currently for the periods 1961-1990 and 1991-2020

The available parameters, the parameter and whether our climatologists perform quality control (QC) for the respective parameters. Click [here](#) to see which resolutions data are available.

Name	Unit	Description	QC performed by DMI's climatologists
mean_temp	°C	Mean temperature	x
mean_daily_max_temp	°C	Mean of daily maximum temperature	x
max_temp_w_date	°C	Maximum temperature with associated date	x
no_ice_days	no. of days	Number of ice days (maximum temperature < 0°C)	x
no_summer_days	no. of days	Number of summer days (maximum temperature > 25°C)	x
mean_daily_min_temp	°C	Mean of daily minimum temperature	x
min_temp	°C	Minimum temperature	x
no_cold_days	no. of days	Number of cold days (minimum temperature < -10 °C)	x
no_frost_days	no. of days	Number of days with frost (minimum temperature < 0°C)	x
no_tropical_nights	no. of days	Number of tropical nights (minimum temperature > 20°C)	x
no_lightning_strikes	no. of strikes	Number of lightning strikes	x
acc_heating_degree_days_17	no. of days	Accumulated heating degree days (17°C - mean_temp)	x
mean_relative_hum	%	Mean relative humidity	x
drought_index		Drought index	x

Name	Unit	Description	QC performed by DMI's climatologists
pot_evaporation_makkink		Potential evaporation (Makkink)	x
mean_wind_speed	m/s	Mean wind speed	x
max_wind_speed_10min	m/s	Maximum wind speed (10 minutes average)	x
max_wind_speed_3sec	m/s	Maximum wind speed (3 seconds average)	x
mean_wind_dir	degrees	Mean wind direction	x
mean_pressure	hPa	Mean pressure	x
bright_sunshine	minutes/hours	Minutes/hours of bright sunshine. For hourly time resolution the unit is minutes, for all other time resolutions the unit is hours.	x
mean_radiation	W/m2 / MJ/m2	Mean radiation (spectral range: 305-2800nm). For hourly time resolution the unit is W/m2, for all other time resolutions the unit is MJ/m2.	x
acc_precip	mm	Accumulated precipitation	x
no_days_acc_precip_01	no. of days	Number of days with accumulated precipitation >= 0.1mm	x
no_days_acc_precip_1	no. of days	Number of days with accumulated precipitation >= 1mm	x
no_days_acc_precip_10	no. of days	Number of days with accumulated precipitation >= 10mm	x
max_precip_30m	mm	Maximum 30 minutes intensity in 24 hours	x
mean_cloud_cover	%	Mean cloud cover	-
snow_depth	cm	Snow depth	x
temp_grass *	°C	Air temperature measured at grass height (5-20 cm over terrain)	-
temp_soil_10 *	°C	Temperature measured at a depth of 10 cm	-
temp_soil_30 *	°C	Temperature measured at a depth of 30 cm	-
leaf_moisture *	minutes	Leave moisture (in danish: bladfugt)	-
vapour_pressure_deficit_mean *		Vapour Pressure Deficit Mean (in danish: mætningsdeficit)	-

* The last 5 parameters in the table above are additional parameters, which are not quality controlled by DMI's climatologists. As a result, data can be erroneous.

** Read more about the different types of days [here](#).

*** Decadal values from 2011-2020 only for municipalities

In the table below, the calculation method for stationValue parameter values is described for the available time resolutions. In case of differences between Denmark and Greenland, these differences can be seen in the table below.

Calculation Method for stationValue Parameters

Name	Hourly	Daily	Monthly and yearly	metObs Parameter data is based on
mean_temp	derived directly (see column 'metObs parameter data is based on')	average of all hourly values within a day	average of all the underlying daily values	DNK: temp_mean_past1h GRL: temp_mean_past1h or alternatively temp_dry at minute 0
mean_daily_max_temp	-	-	average of all the underlying daily values of max_temp_w_date	-
max_temp_w_date	derived directly (see column 'metObs parameter data is based on')	DNK: The highest of all the hourly values in the course of the day GRL: The highest of all the hourly values in the course of the day. If max_temp_w_date is not available max_temp_12h is used. If neither is available the highest value of mean_temp is used	the highest of the underlying daily values if more than one identical highest value is found, the date is found as follows: Jan-Jun: first found date Jul-Dec: last found date	temp_max_past 1h
max_temp_12h	derived directly (see column 'metObs parameter data is based on')	-	-	temp_max_past12h
no_ice_days	-	-	number of days where max_temp_w_date is below 0°C	-
no_summer_days	-	-	number of days where max_temp_w_date is more than 25°C	-
mean_daily_min_temp	-	-	average of the underlying daily values of min_temp	-

Calculation Method for stationValue Parameters

Name	Hourly	Daily	Monthly and yearly	metObs Parameter data is based on
min_temp	derived directly (see column 'metObs parameter data is based on')	DNK: the lowest of all hourly values of min_temp in the course of a day GRL: the lowest of all hourly values of min_temp in the course of the day. If min_temp is not available min_temperature_12h is used. If neither is available, the lowest value of mean_temp is used	the lowest of the underlying daily values if more than one identical lowest value is found, the date is found as follows: Jan-Jun: last found date Jul-Dec: the first found date	temp_min_past1h
min_temperature_12h	derived directly (see column 'metObs parameter data is based on')	-	-	temp_min_past12h
no_cold_days	-	-	number of days where min_temp is less than -10°C	-
no_frost_days	-	-	number of days where min_temp is less than 0°C	-
no_tropical_nights	-	-	number of days where min_temp is more than 20°C	-
acc_heating_degree_days_17	-	calculated as follows: 17°C minus the daily value of mean_temp	sum of the underlying daily values	-
acc_heating_degree_days_19	-	calculated as follows: 19°C minus the daily value of mean_temp	sum of the underlying daily values	-

Calculation Method for stationValue Parameters

Name	Hourly	Daily	Monthly and yearly	metObs Parameter data is based on
mean_relative_hum	DNK: average of the 6 underlying values within the the hour GRL: derived directly (see column 'metObs parameter data is based on')	average of all hourly values within a day	average of the underlying daily values	DNK: humidity GRL: humidity_past1h or alternatively humidity at minute 0
max_relative_hum	-	the highest of all hourly values of mean_relative_hum within a day	the highest of all the daily values	-
min_relative_hum	-	the lowest of all hourly values of mean_relative_hum within a day	the lowest of all the daily values	-
mean_vapour_pressure	-	see description of calculation method below	average of the underlying daily values	-
mean_wind_speed	average of the 6 underlying values within the hour. For GRL: Value is not calculated correctly before 2024.	average of all hourly values within a day	average of the underlying daily values	wind_speed

Calculation Method for stationValue Parameters

Name	Hourly	Daily	Monthly and yearly	metObs Parameter data is based on
max_wind_speed_10min	the highest of the 6 underlying values within the hour of the parameters in the column 'metObs parameter data is based on'	the highest of all hourly values within a day	the highest of all of the underlying daily values. If more than one identical highest values is found, the date is found as the latest date	wind_speed & wind_max_per10min_past1h
max_wind_speed_3sec	DNK: see description of calculation method below GRL: derived directly (see column 'metObs parameter data is based on')	the highest of all hourly values within a day	the highest of all the underlying daily values. If more than one identical highest values is found, the date is found as the latest date	DNK: wind_gust_always_past1h, wind_max, wind_gust_past10min_wind_gust_last1h, wind_speed GRL: wind_gust_always_past1h or alternatively wind_max at minute 0
no_windy_days	-	-	number of days where max_win_speed_10min is greater or equal to 10.8 m/s	-
no_stormy_days	-	-	number of days where max_win_speed_10min is greater or equal to 20.8 m/s	-
no_days_w_storm	-	-	number of days where max_win_speed_10min is greater or equal to 24.5 m/s	-
no_days_w_hurricane	-	-	number of days where max_win_speed_10min is greater or equal to 28.5 m/s	-

Calculation Method for stationValue Parameters

Name	Hourly	Daily	Monthly and yearly	metObs Parameter data is based on
mean_wind_dir_min0	derived directly (see column 'metObs parameter data is based on')	-	-	wind_dir at minute 0
mean_wind_dir	derived directly (see column 'metObs parameter data is based on')	mean of components	mean of components	DNK: wind_dir_past1h GRL: wind_dir_past1h or alternatively wind_dir at minute 0
mean_pressure	DNK: average of the 6 underlying values within the hour GRL: derived directly (see column 'metObs parameter data is based on')	average of all hourly values within a day	average of the underlying daily values	DNK: pressure_at_sea GRL: pressure_at_sea at minute 0
max_pressure	-	highest hourly value of mean_pressure within a day	the highest of all the underlying daily values	-
min_pressure	-	lowest hourly value of mean_pressure within a day	the lowest of all the underlying daily values	-
bright_sunshine	derived directly (see column 'metObs parameter data is based on')	sum of all underlying hourly values within a day, where minutes is converted to hours by dividing the sum of hourly values with 60	sum of the underlying daily values	sun_last1h_glob

Calculation Method for stationValue Parameters

Name	Hourly	Daily	Monthly and yearly	metObs Parameter data is based on
mean_radiation	derived directly (see column 'metObs parameter data is based on')	sum of all underlying hourly values of bright_sunshine within a day. In order to convert to MJ/m ² , the sum is multiplied by (60*60/10 ⁶)	sum of the underlying daily values	radia_glob_past1h
acc_precip	derived directly (see column 'metObs parameter data is based on')	sum of all underlying hourly values within a day	sum of the underlying daily values	precip_past1h
max_precip_24h	-	-	the highest of the underlying daily values of acc_precip. If more than one identical highest values is found, the date is found as the latest date	
acc_precip_past12h	derived directly (see column 'metObs parameter data is based on')	-	-	precip__past12h
no_days_acc_precip_01	-	-	number of days where acc_precip is greater or equal to 0.1 mm	
no_days_acc_precip_1	-	-	number of days where acc_precip is greater or equal to 1 mm	
no_days_acc_precip_10	-	-	number of days where acc_precip is greater or equal to 10 mm	
acc_precip_past24h	derived directly (see column 'metObs parameter data is based on')	-	-	precip_past24h

Calculation Method for stationValue Parameters

Name	Hourly	Daily	Monthly and yearly	metObs Parameter data is based on
max_precip_30m	-	is found by inspecting the period from 23:31 local time to the previous day to 24.00 local time of the respective day	the highest of the underlying daily values of acc_precip. If more than one identical highest value is found, the date is found as the latest date	
no_days_snow_cover	-	-	number of days where snow_depth is greater than 0 and snow_cover is greater than 50	
mean_cloud_cover	derived directly (see column 'metObs parameter data is based on')	average of the underlying hourly values provided that there has to be at least 12 hourly values and there are not more than 2 hourly values missing in a row	average of the underlying daily values	cloud_cover at minute 0
no_clear_days	-	= 1 if mean_cloud_cover is less than 20% = 0 if mean_cloud_cover is greater than or equal to 20%	number of days where mean_cloud_cover is less than 20%	
no_cloudy_days	-	=1 if mean_cloud_cover is greater than 80% = 0 if mean_cloud_cover is less than or equal to 80%	number of days where mean_cloud_cover is greater than 80%	
max_snow_depth	-	-	the highest of the underlying daily values of snow_depth. If more than one identical highest values are found, the date is found as the latest date	
snow_depth	-	derived directly (see column 'metObs parameter data is based on'), but code -1 is changed to 0,5 cm and code -2 is changed to 0 cm	-	snow_depth_man

Calculation Method for stationValue Parameters

Name	Hourly	Daily	Monthly and yearly	metObs Parameter data is based on
snow_cover	-	derived directly (see column 'metObs parameter data is based on'), but codes are changed to %	-	snow_cover_man
temp_grass	derived directly (see column 'metObs parameter data is based on')	average of all hourly values within a day	-	temp_grass_mean_past1h
temp_soil_10	derived directly (see column 'metObs parameter data is based on')	average of all hourly values within a day	-	temp_soil_mean_past1h
temp_soil_30	derived directly	average of all hourly values within a day	-	N/A
leave_moisture	derived directly (see column 'metObs parameter data is based on')	average of all hourly values within a day	-	leav_hum_dur_past1h
vapour_pressure_deficit_mean	see description of calculation method below	average of all hourly values within a day	-	temp_mean_past1h and humidity

The mean_vapour_pressure is calculated from the hourly values of mean_temp and mean_relative_hum. For every of the days hourly value, where temperature (tt) and relative humidity (rh) is available, the mean_vapour_pressure (vp) is calculated using the following formula:

If $tt = 0$, then $vp = 6.107 * rh / 100$

If $tt < 0$, then

$$a = 9.5 \quad b = 265.5 \quad q = a * tt / (tt + b) \quad vp = 6.107 * 10^q * rh / 100$$

$$\text{If } tt > 0, \text{ then } a = 7.5 \quad b = 237.3 \quad q = a * tt / (tt + b) \quad vp = 6.107 * 10^q * rh / 100$$

The daily value of mean_vapour_pressure is calculated as a mean of the hourly values, which could be calculated within the day.

max_wind_speed_3sec is found in a 24 hour time interval from 00 to 24 local. In this interval the observation at 00:00 is not included in the calculation, but the observation at 24:00 is included in the calculation.

Time period	24 hours (00:00-24:00) local time
Examination period	24 hours (00:00-24:00) local time
Condition	In order for at station to be included in the calculation of max_wind_speed_3sec, the number of 10 minute wind speeds (wind_speed) in the time period must be greater than 140
Output	For every grid cell the highest max:_wind_speed_3sec
Calculation	The calculation is based on station values, so that the highest 3-second wind is found for each station. There are several parameters which determine the 3-second wind speed, where the following priority is used: 1. wind_max or wind_gust_always_past1h 2. wind_max or wind_gust_past10min or wind_max or wind_gust_last1h If none of the above parameters with first or second priority are available for the station in the desired time interval, the max_wind_speed_3sec is estimated from the highest wind_max or wind_speed the time interval.

The vapour_pressure_deficit_mean is calculated from the hourly values of mean_temp and mean_relative_hum. The daily value of mean_vapour_pressure is calculated as a mean of the hourly values, which could be calculated within the day.

If mean_temp > 0 then,

$$\text{SatVapPres} = 0.611 * \text{Math.exp} (17.27 * \text{mean_temp} / (\text{mean_temp} + 237.3))$$

If mean_temp <= 0 then,

$$\text{SatVapPres} = 0.611 * \text{Math.exp}(21.87 * \text{mean_temp} / (\text{mean_temp} + 265.5))$$

$$\text{VPD} = \text{SatVapPres} * (100 - \text{mean_relative_hum}) / 100$$

The climateData service includes data in different spatial resolutions for Denmark.

Spatial resolution	Description
10kmGridValue	climate data in a 10 x 10km grid resolution covering danish land areas
20kmGridValue	climate data in a 20 x 20km grid resolution covering danish land areas
municipalityValue	climate data for the Danish municipalities
countryValue	climate data for Denmark

The data is based on station data from Danish stations, which is interpolated into a 1x1km grid net covering all of Denmark's land area using a modified inverse-distance algorithm. The grid is based on [Det Danske Kvadratnet](#)'s (external site) 1x1km grid with the grid name DKN_1km_ETRS89. The calculated values in each grid cell depends on the values of the nearest surrounding station found in 8 sectors. One value is calculated for each area (e.g. one municipality) from the grid cells covering the jurisdiction area combined with a weighting of how much of each grid cell is within the area. The stations are weighted in relation to the distance and climatological comparability (distance to sea) to the grid cell. Aggregated grid cells are used to compile datasets of different spatial resolution.

The different spatial resolutions are calculated/derived from stationValue data as it becomes available. The data is recalculated when the quality controlled stationValue data has been quality controlled by our climatologists. In some instances area data is based on additional parameters, which are not quality controlled. Please see the [parameter list](#) for an overview over which parameters are available and whether or not they are quality controlled by our climatologists.

Spatial resolutions are not available for Greenland, as an interpolation algorithm cannot be applied due to the complicated circumstances of the large geographical area, complex topography and low station density.

The stationValue dataset is used to create a 1x1 km grid net covering all of Denmark's land area. The grid net is calculated from a modified inverse-distance algorithm and is based on all stations that DMI has at its disposal, including both DMI and third party

stations. In order to understand the algorithm one must consider the local Danish climate. The overall factor that generally spoken has the largest influence on the local climate in Denmark is the distance to the sea. The uneven station coverage of Denmark's area represents a challenge if using a classic non-weighted interpolation. Areas with bad coverage are in risk of being affected by stations located in an area that climatologically seen is not representative for the point of interpolation. Therefore a modified inverse-distance algorithm is applied, where the value in each grid cell depends on the values of the nearest surrounding stations found in 8 sectors. The stations are weighted in relation to the distance and climatological comparability (distance to sea) to the grid.

The coast-/inland climate distribution follows figure 1 below:

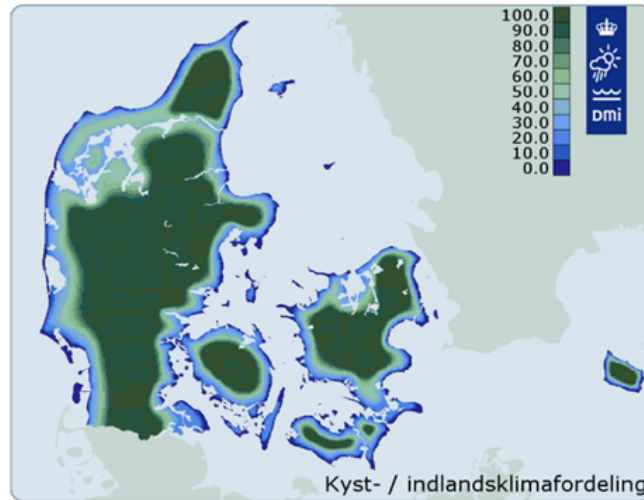


Figure 1. Distribution of coast-/inland climate

Each grid cell value is calculated from weighted station values. A grid cell is not represented by every station equally. Station selection is needed since some stations may be located in a different climate zone or add undesired effects to the grid cell value. The algorithm selects the nearest stations in 8 sectors.

In the following, the station selection method is illustrated for the case shown in Figure 2. The green triangles illustrate stations and the red square is the grid cell to be calculated. The algorithm creates four lines passing through the center of the grid cell, creating eight sectors (ENE, NNE, NNW, WNW, WSW, SSW, SSE, ESE) and finds the nearest station in each sector - aquamarine triangles in Figure 2 right panel.

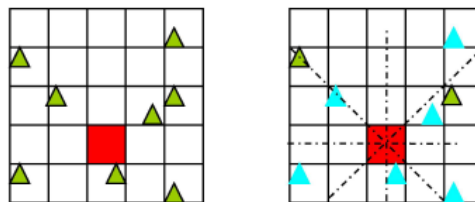


Figure 2. Illustration of the station selection method

The calculated values in each grid cell depend on the values of the nearest surrounding station found in 8 sectors. The stations are weighted in relation to the distance and climatological comparability (distance to sea) to the grid cell. Interpolated values will never exceed the values of the original data the grid network was calculated from. This ensures that "false" records will never appear from the interpolation. This is done for all 1x1km cells covering Denmark's land area. The 1x1km grid is the calculation basis for all of the spatial resolutions described below. The observation network is embedded into this dataset, i.e. the interpolation algorithm is made so that the 1x1 km grid cell that geographically contains a station, always will have the observed value. The 1x1 km grid net serves as the basis of calculation for the other spatial resolutions.

The 1x1km grid is then used to calculate a single value for each spatial resolution (e.g. one municipality) from the grid cells covering the specified area combined with a weighting of how much of each grid cell is within the area. Below is shown an example of the 1x1 km grid covering Holbæk Municipality (Holbæk Kommune):

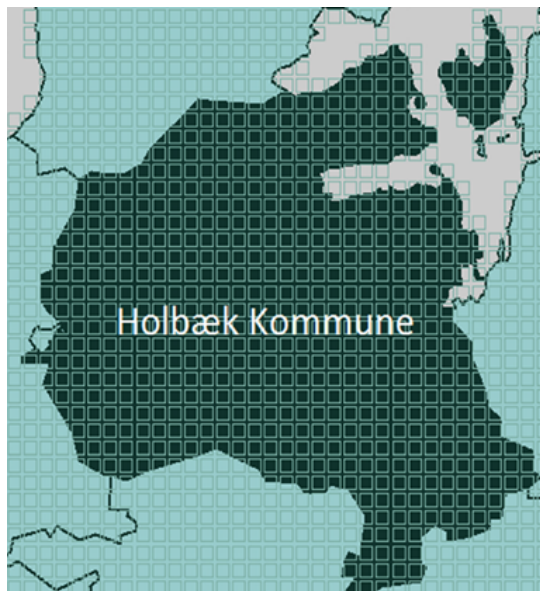


Figure 3. 1x1 km grid covering Holbæk Municipality

Aggregated 1x1km grid cells are used to compile datasets of different spatial resolution:

Spatial resolution

Description

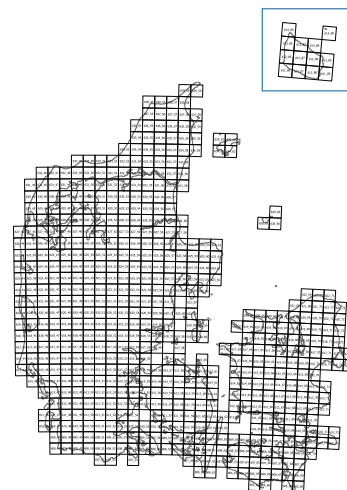
10kmGridValue

Climate data in a 10 x 10km grid resolution covering Danish land areas. This matches the DKN_10km_ETRS89 grid from [Det Danske Kvadratnet](#).

For your convenience DMI has created an [interactive web page map to inspect the grid](#) (external site).

You can also see, how you can import [grid identification layers into QGIS](#).

10 km grid cells are also available as pdf [here](#).



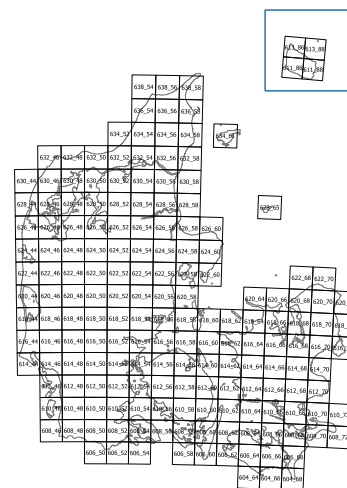
20kmGridValue

Climate data in a 10 x 10km grid resolution covering Danish land areas. This matches the DKN_10km_ETRS89 grid from [Det Danske Kvadratnet](#).

For your convenience DMI has created an [interactive web page map to inspect the grid](#) (external site).

You can also see, how you can import [grid identification layers into QGIS](#).

20 km grid cells are also available as pdf [here](#).



municipalityValue climate data for the Danish municipalities

Spatial resolution	Description
countryValue	climate data for Denmark

For a detailed description of the interpolation, see [DMI Technical Report 10-13](#) (the report is in Danish).

Data is made available through DMI's open data service as soon as it has been calculated, but not yet quality controlled. When the stationValue data has been quality controlled by our climatologists, the data is recalculated and will be updated in the service with any changes to data. In some instances, the data is based on additional parameters, which are not quality controlled. These parameters are agricultural parameters.

Climate data is not available for Greenland in different spatial resolutions, as an interpolation algorithm cannot be applied due to the complicated circumstances of the large geographical area, complex topography and low station density.

It is possible to get the station list through the climateData service by calling the station endpoint. See [API documentation](#) for endpoint.

The station list represents both present and historical data about a station - see [Station Lists Explained](#).

List is also [available here](#).

[Click here](#) for page displaying the historical availability for each parameter at each station. The noted year is the first year with observations for the given parameter and station.

The DMI has five types of stations for which measure meteorological parameters. A description of the different station types can be seen below.

Station type	Description	Available for Denmark	Available for Greenland
Synop	Weather stations that register a wide variety of meteorological parameters, e.g. temperature, wind, pressure, and precipitation.	X	X
GIWS	Weather stations that register a wide variety of meteorological parameters, e.g. temperature, wind, and pressure.		X
Pluvio	Precipitation stations register data concerning precipitation. Precipitation is also measured by some synop-stations.	X	
Manual precipitation	In Greenland the precipitation is at some locations measured manually once a day. Once a month DMI receives the data, whereafter the data will be available through our API-services. There can be some instances where precipitation has not been measured daily for different reasons. There can therefore be instances, where a measurement can cover the accumulated precipitation of more than one day. It is possible to see if a measurement covers more than one day's accumulated precipitation, if there are not any data for the previous day(s).		X
Manual snow	In Denmark the snow depth and cover is measured manually once a day.	X	